

Note

Lab05 Exercise

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Outline

- Overview
- Input data
- Input action
- Action execution
- Output
- Definition of clock



Overview

One pattern has 8 sets of action
One set of action contains 2 ~ 8 actions

- Flow

Flow	Pattern 1	Set1		Set2			Set 8		Pattern 2	Set1		
Behavior	Input data	Input action	Output	Input action	Output	...	Input action	Output	Input data	Input action	Output	...
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	...	action	result matrix (serial out)	image template image_size	action	result matrix (serial out)	...

2~4 negedge clk

2~4 negedge clk

2~4 negedge clk

Asynchronous reset

Max latency: 5000



Input data (1/3)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	

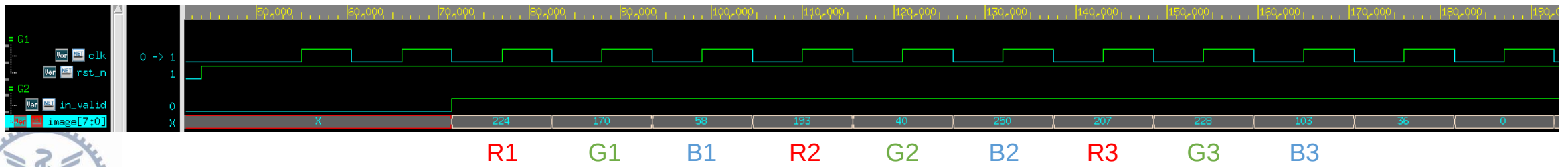


- image (8-bit unsigned)
 - R, G and B value of one element will be given sequentially in 3 cycles

R1	R2		
			R16

G1	G2		
			G16

B1	B2		
			B16

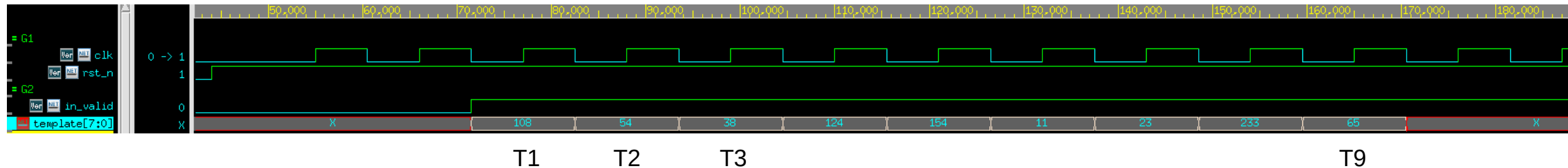


Input data (2/3)

- template (8-bit unsigned)
 - It will be given in the first 9 cycles

T1	T2	T3
T4	T5	T6
T7	T8	T9

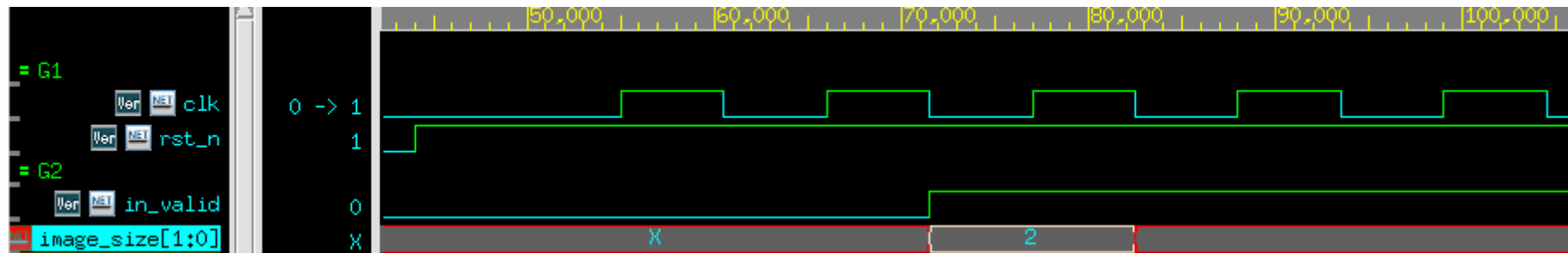
Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



Input data (3/3)

- image_size (2-bit unsigned)
 - It will be given at the first cycle

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



Input action

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	

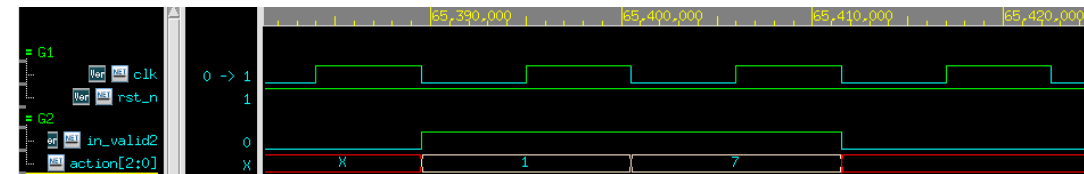


- Action format (called a set of action)
 - The first action must be 0 ~ 2
 - The last action must be 7
 - Other actions must be 3 to 6 (may occur repeatedly)
 - The number of one set of action will be range from 2 to 8

First action	0~6 actions	Last action
0~2	3~6	7

Example 1 for a set of action that contains 2 actions
1 -> 7

Example 2 for a set of action that contains 8 actions
1 -> 3 -> 4 -> 6 -> 5 -> 5 -> 4 -> 7

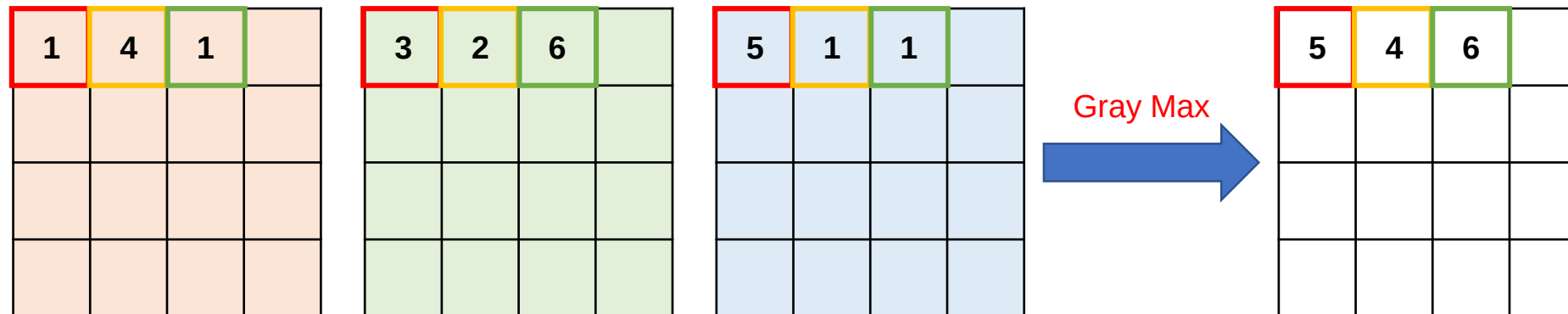


Action execution (1/8)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- Grayscale Transformation (Maximum method)
 - $\text{Grayscale}(x, y) = \max\{ R(x, y), G(x, y), B(x, y) \}$



$$\max\{1, 3, 5\} = 5$$

$$\max\{4, 2, 1\} = 4$$

$$\max\{1, 6, 1\} = 6$$



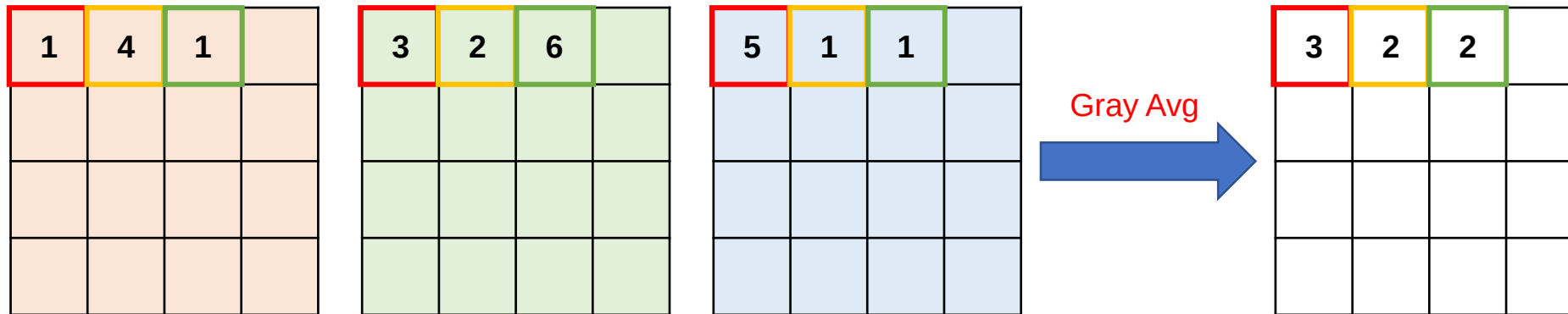
Action execution (2/8)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- Grayscale Transformation (Average method)

- $$\text{Grayscale}(x, y) = \{ R(x, y) + G(x, y) + B(x, y) \} / 3$$
round-down!



$$(1+3+5) / 3 = 9/3 = 3$$

$$(4+2+1) / 3 = 7/3 = 2$$

$$(1+6+1) / 3 = 8/3 = 2$$



Action execution (3/8)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- Grayscale Transformation (Weighted method)
 - $\text{Grayscale}(x, y) = R(x, y) / 4 + G(x, y) / 2 + B(x, y) / 4$ **round-down!**

1	4	1	

3	2	6	

5	1	1	



2	2	3	

$$1/4 + 3/2 + 5/4 \\ = 0 + 1 + 1 = 2$$

$$4/4 + 2/2 + 1/4 \\ = 1 + 1 + 0 = 2$$

$$1/4 + 6/2 + 1/4 \\ = 0 + 3 + 0 = 3$$



Action execution (4/8)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- Max pooling
 - This operation will be performed only when the image size is 8×8 or 16×16
 - If the image size is 4×4 , the result of max pooling will be the same as the original matrix.

3	2	7	5	4	5	5	1
6	1	1	2	1	3	4	5
3	4	6	3	5	0	5	0
5	5	2	0	4	2	4	4
4	7	5	3	2	3	3	5
3	4	5	1	3	3	4	3
3	1	3	2	2	4	6	4
3	4	5	3	2	5	2	5

Max pooling



6	7	5	5
5	6	5	5
7	5	3	5
4	5	5	6

6	7	5	5
5	6	5	5
7	5	3	5
4	5	5	6

Max pooling



6	7	5	5
5	6	5	5
7	5	3	5
4	5	5	6

(No value will be changed!)

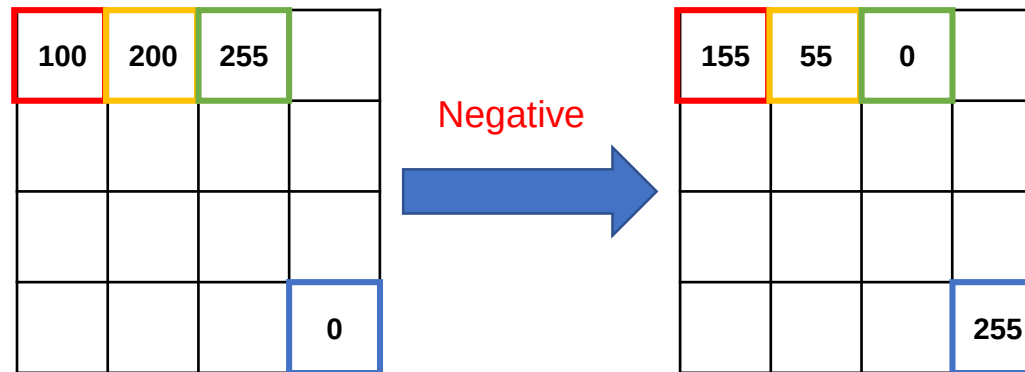


Action execution (5/8)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



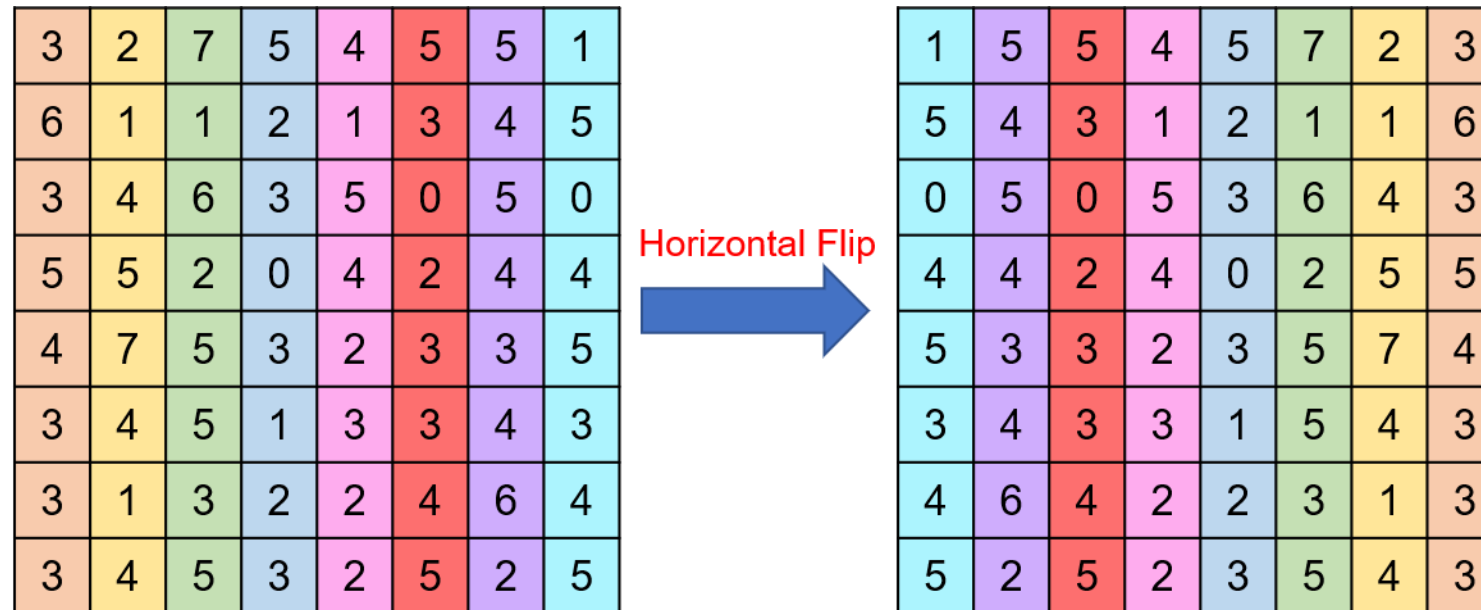
- Negative
 - $\text{Negative}(x, y) = 255 - \text{Grayscale}(x, y)$



Action execution (6/8)

- Horizontal Flip

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



Action execution (7/8)

- Image Filter

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



3	3	2	7	5	4	5	5	1	1
3	3	2	7	5	4	5	5	1	1
6	6	1	1	2	1	3	4	5	5
3	3	4	6	3	5	0	5	0	0
5	5	5	2	0	4	2	4	4	4
4	4	7	5	3	2	3	3	5	5
3	3	4	5	1	3	3	4	3	3
3	3	1	3	2	2	4	6	4	4
3	3	4	5	3	2	5	2	5	5
3	3	4	5	3	2	5	2	5	5

Image with replication padding

Image Filter



3	3	2	4	4	4	5	4
3	3	3	4	3	4	4	4
5	4	2	2	2	4	4	4
4	5	4	3	3	3	3	4
4	5	4	3	3	3	3	4
3	4	3	3	3	3	4	4
3	3	3	3	3	3	4	4
3	3	3	3	3	2	5	5

Filtered image

Median



1, 2, 2, 3, 3, 3, 3, 6, 6
 1, 1, 2, 2, 3, 3, 6, 7, 7
 1, 1, 2, 2, 2, 5, 5, 7, 7
 2, 2, 4, 4, 5, 5, 5, 5, 6



Action execution (8/8)

- Cross correlation

0	0	0	0	0	0	0	0	0	0
0	3	2	7	5	4	5	5	1	0
0	6	1	1	2	1	3	4	5	0
0	3	4	6	3	5	0	5	0	0
0	5	5	2	0	4	2	4	4	0
0	4	7	5	3	2	3	3	5	0
0	3	4	5	1	3	3	4	3	0
0	3	1	3	2	2	4	6	4	0
0	3	4	5	3	2	5	2	5	0
0	0	0	0	0	0	0	0	0	0

Image with zero padding

Cross Correlation



3	7	4
4	0	1
6	6	3

template

41	64	31	53	52	57	78	74
60	136	150	146	122	120	101	68
95	113	79	74	67	113	105	115
87	164	167	125	85	94	98	79
92	136	117	89	84	101	118	94
81	131	115	103	78	109	137	120
68	129	120	102	93	108	126	99
29	45	51	53	53	68	95	54

Image after cross correlation

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	

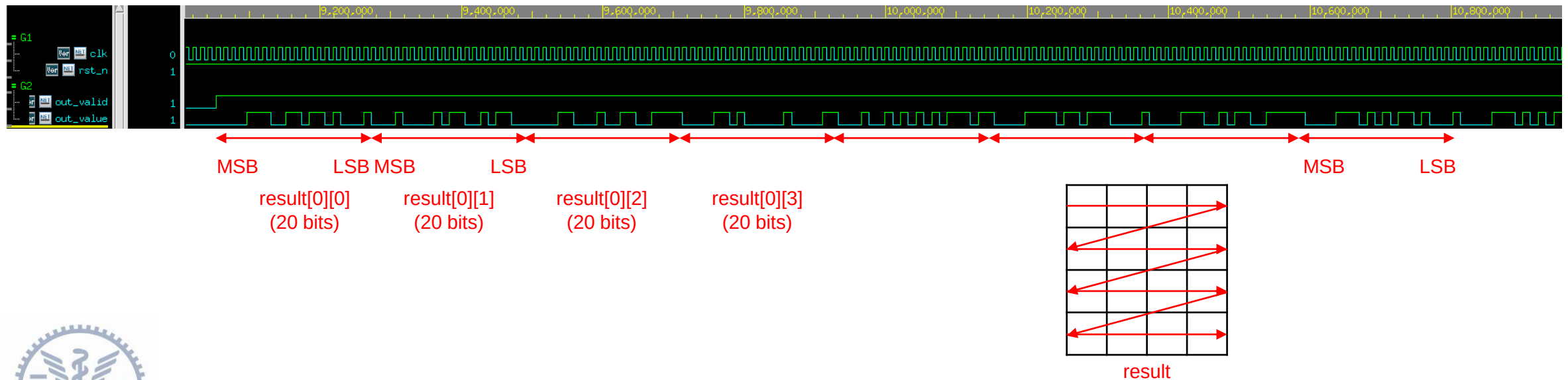


Output

Flow	Pattern 1	Set1		Set2		
Behavior	Input data	Input action	Output	Input action	Output	...
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- Output the result matrix in raster scan order with serial out format from the MSB.
- One element of result matrix will be outputted for 20 cycles

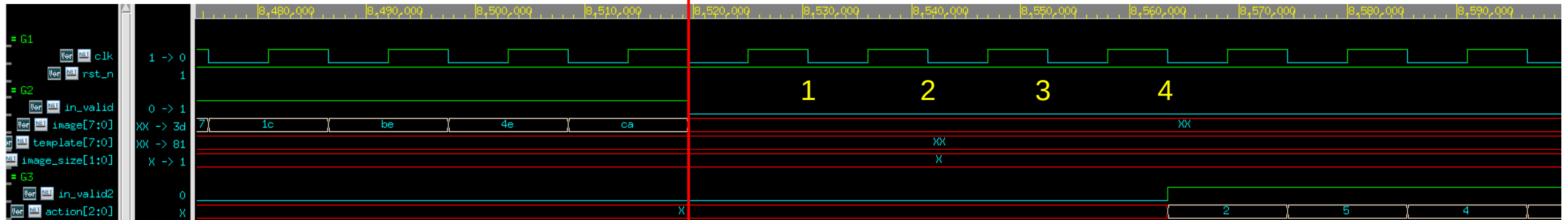


Definition of clock (1/4)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- in_valid2 will come in 2~4 negative edge of clock after in_valid is tied low

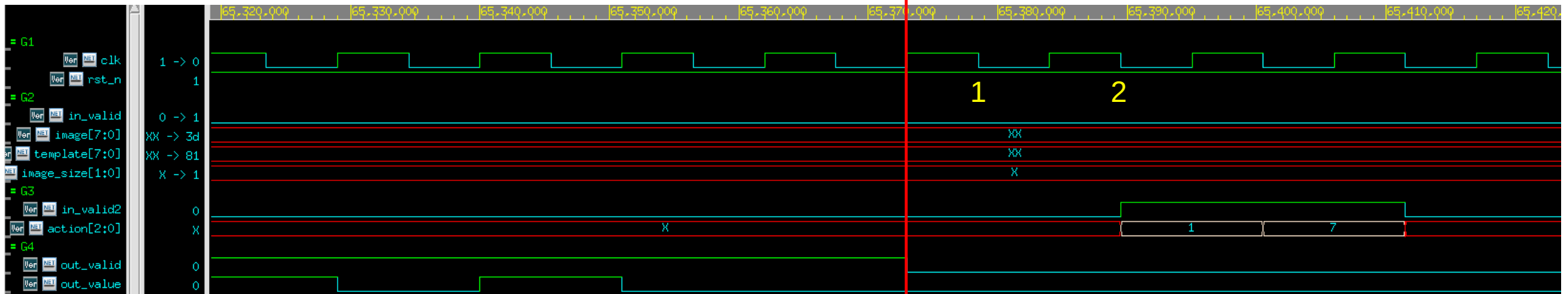


Definition of clock (2/4)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- The other 7 times in_valid2 will come in 2~4 negative edge of clock after out_valid falls

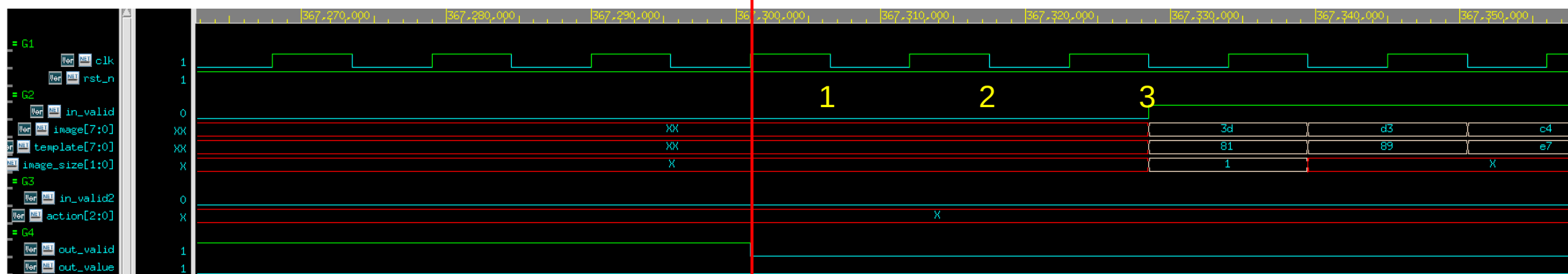


Definition of clock (3/4)

Set2		...	Set 8		Pattern 2
Input action	Output		Input action	Output	Input data
action	result matrix (serial out)		action	result matrix (serial out)	image template image_size



- The next input pattern will come in 2~4 negative edge of clock after the eighth out_valid of this pattern falls



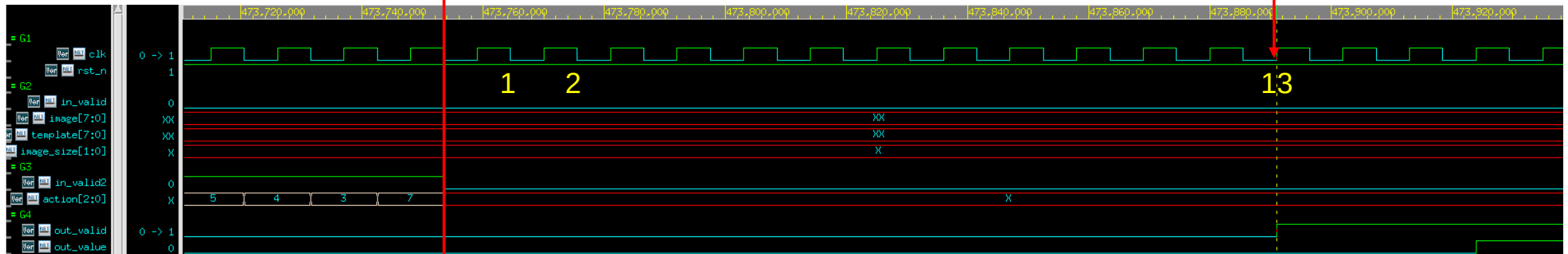
Definition of clock (4/4)

Flow	Pattern 1	Set1		Set2		...
Behavior	Input data	Input action	Output	Input action	Output	
Objects	image template image_size	action	result matrix (serial out)	action	result matrix (serial out)	



- The definition of latency

The maximum of this value cannot be larger than 5000!



Thanks

